

Dondapati Karthikeya

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Generative AI Engineer — LLMs — RAG — Agentic AI — Python

Final-year B.Tech Computer Science (AI & ML) student with strong foundations in Generative AI, Natural Language Processing (NLP), and Machine Learning. Hands-on experience building LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and AI agent workflows. Proficient in Python with practical exposure to model integration, inference pipelines, and performance optimization. Passionate about developing intelligent, context-aware AI systems and rapidly learning emerging AI technologies.

Technical Skills

- **Programming:** Python, SQL
- **Generative AI & NLP:** Large Language Models (LLMs), RAG, Prompt Engineering
- **AI Frameworks:** Hugging Face Transformers
- **Machine Learning:** Supervised Learning, Feature Engineering, Model Evaluation
- **Agentic AI:** AI Agents, Multi-Agent Workflows, Multimodal RAG
- **Backend & APIs:** FastAPI, Flask, REST APIs,
- **Vector & Retrieval Concepts:** Embeddings, Semantic Retrieval, Context Handling
- **Software Engineering:** OOP, SDLC, Debugging, Git
- **Visualization:** Power BI

Work Experience

Machine Learning Intern

Jan 2025 – May 2025

DRdotSolution (Remote)

- Contributed to the design, development, testing, and optimization of AI-driven applications.
- Developed Python-based inference components for an Automatic Number Plate Recognition (ANPR) system.
- Designed and maintained REST API integrations for model inference and data workflows.
- Debugged and optimized model inference pipelines, improving processing efficiency by 35%.
- Performed validation and reliability testing to ensure stable AI system behavior.
- Collaborated with cross-functional teams to troubleshoot issues and refine system performance.
- Produced technical documentation describing model workflows and system logic.

Selected Projects

GenAI-Powered Medical Diagnosis System

- Designed and implemented an end-to-end LLM-powered application using Retrieval-Augmented Generation (RAG).
- Built document processing, embedding generation, and semantic retrieval pipelines.
- Improved response relevance through structured context injection and retrieval optimization.
- Developed modular workflows for scalable and maintainable AI system design.

AI Agent-Based Travel Planner

- Developed an agentic AI system simulating autonomous decision-making and reasoning workflows.
- Implemented context-aware retrieval mechanisms to enhance recommendation consistency.
- Optimized logic flows to improve response coherence and system responsiveness.

End-to-End Cricket Data Analytics (Python & Power BI)

- Built Python pipelines for large-scale data collection, cleaning, and transformation.
- Improved dataset quality by 70% using preprocessing and validation techniques.
- Delivered analytical dashboards enabling automated insight generation.

Education

B.Tech in Computer Science Engineering (AI & ML)

2022 – 2026

Mallareddy University, Hyderabad

CGPA: 8.68

Certifications

Deloitte Data Analytics Virtual Experience

Accenture Data Analytics Virtual Experience